Credit Card Fraud Detection Machine Learning Pipeline

Fintech Case Study · MSc Analytics & Management · London Business School
GitHub: https://github.com/andreeagalusca234/credit-card-fraud-detection-ml

PROBLEM

Credit card fraud costs the global financial industry over $30 billion annually. Traditional rule-based systems flag too many false positives, frustrating customers and creating noise for fraud teams. The challenge is building a model that catches as many fraudulent transactions as possible while minimising false alarms — on a dataset where only 0.17% of transactions are fraudulent.

APPROACH

- Applied SMOTE oversampling to address extreme class imbalance (0.17% fraud rate) — without this, standard models learn nothing useful about fraud
- Trained and compared 3 ML models: Logistic Regression (baseline), Random Forest, and XGBoost on PCA-transformed transaction features
- Evaluated using ROC-AUC and Precision-Recall AUC — chosen over accuracy because standard accuracy is misleading on imbalanced data
- Designed a tiered risk decision engine that translates probability scores into Approve / Review / Decline actions, mirroring production fraud systems

KEY INSIGHTS

01 IMBALANCE

SMOTE is non-negotiable

Without oversampling, recall on the fraud class drops sharply. Standard accuracy of 99.8% means nothing — the model must be evaluated on Precision-Recall AUC.

02 SIGNAL

V-features dominate

PCA features V4, V11, V14, and V17 emerged as the strongest predictors. Engineered features (log-amount, time-of-day) added further signal.

03 STRATEGY

Risk scoring = business control

Rather than a binary output, the model produces a continuous risk score. Threshold tuning is a business decision — trading off fraud capture vs. customer friction.

BUSINESS APPLICATION (FINTECH & FRAUD OPERATIONS)

This pipeline applies directly to fintech fraud teams where decisions must be made at scale, in real time, on highly imbalanced data. A company like Monzo, Revolut, or Visa could apply this framework to:

- Detect and block fraudulent transactions in real time before settlement
- Tune risk thresholds by geography, product, or customer segment to balance fraud capture vs. false positive rate
- Reduce manual review queue size by routing only genuinely ambiguous transactions to human analysts
- Build model monitoring pipelines that detect fraud pattern drift as criminal behaviour evolves

TOOLS

Python · scikit-learn · XGBoost
Pandas · NumPy · imbalanced-learn
Matplotlib · Seaborn

BUSINESS IMPACT

ROC-AUC ~0.98 on held-out test data
~86% of fraud cases caught (recall)
Protects ~£73,000/day at scale · fully reproducible